Eco Bitcoin

A Proof of Participation, VRF-Driven, Virtually Mined E-Cash Experiment

Solana Edition

www.ebtc.fredenergy.org

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Abstract

Bitcoin is the standard by which almost every cryptocurrency is eventually measured, compared, criticised or enthusiastically declared to be “the next Bitcoin”.

We are making no such claim.

Bitcoin demonstrated that digital scarcity could exist without a central issuer and introduced a system where participants expend computing power and electricity in return for the opportunity to receive newly issued coins.

Eco Bitcoin, or eBTC, takes a somewhat different approach.

Rather than asking participants to purchase increasingly powerful computers, find cheap electricity and convert a significant quantity of that electricity into heat, eBTC experiments with virtual mining through Proof of Participation (POP).

Participation is represented by a limited-supply NFT on the Solana blockchain.

Ownership of an eligible eBTC NFT provides the holder with entry into cryptographically random eBTC reward draws. Different NFT tiers provide different draw weights and reward multipliers, while an optional Boost mechanism introduces an additional element of strategy and gamification.

Randomness is provided using a Verifiable Random Function (VRF) mechanism rather than relying upon predictable blockchain information or somebody at FRED Energy rolling a very large pair of dice.

The sale of new POP NFTs also contributes directly to the eBTC market. A programmed proportion of each NFT sale is used to purchase eBTC, with the purchased eBTC and remaining SOL subsequently paired and deposited into the SOL/eBTC liquidity pool.

Liquidity is controlled by the program rather than by an externally controlled project wallet. The intention is simple: liquidity created through NFT sales belongs to the protocol and is not something that can subsequently be withdrawn because somebody has discovered that the Bahamas looks nice at this time of year.

Eco Bitcoin remains an experiment.

It is not Bitcoin.

It is not intended to replace Bitcoin.

It is an attempt to explore whether some of the concepts of mining, scarcity, participation and reward can be reproduced without requiring a warehouse full of ASIC miners.

1. Introduction

Satoshi Nakamoto described Bitcoin issuance by comparing it with gold mining: resources are expended in order to introduce new units into circulation.

In Bitcoin's case those resources are principally specialised computing hardware and electricity.

The system has proved extraordinarily successful.

It has also resulted in an industrial-scale global mining industry.

FRED Energy has always been interested in the slightly awkward intersection between blockchain technology and energy use. While enormous quantities of electricity are consumed securing proof-of-work blockchains, our Sol-bank project has focused on deploying small-home solar systems and energy technology within underserved communities.

Eco Bitcoin approaches the same contradiction from another direction.

What if the idea of mining could remain, while the physical mining machinery disappeared?

No racks of ASICs.

No cooling warehouse, no battle to find the cheapest electricity tariff, and no pretending the spare bedroom is still a spare bedroom after installing twelve graphics cards in it.

Instead, Eco Bitcoin uses NFTs as virtual mining licences.

Proof of Participation — POP.

2. Proof of Participation

Proof of Participation is the foundation of the eBTC reward system.

An eligible eBTC NFT acts as the holder's Proof of Participation.

The NFT does not perform calculations.

It does not solve SHA-256 hashes.

It does not become noticeably warmer when the market goes up.

Instead, ownership makes the NFT eligible to participate in eBTC reward draws.

Each NFT belongs to one of five tiers:

- FX100
- FX250
- GFX500
- GFX750
- GFX808

Draw Weight determines the NFT's relative probability of being selected in a reward draw.

Reward Multiplier determines the reward level associated with that NFT tier when it wins.

Higher-tier NFTs therefore provide both greater weighting within the draw and a higher reward multiplier.

POP is consequently not Proof of Work.

It is not Proof of Stake.

It is simply proof that an eligible participant owns one of the limited eBTC POP NFTs.

That seemed sufficiently different to deserve another acronym.

3. Solana

The original Eco Bitcoin experiment began on the Stellar network.

The project has subsequently moved to Solana.

Solana provides the blockchain infrastructure for:

- eBTC
- POP NFTs
- ownership verification
- reward draw requests
- VRF/randomness integration
- reward settlement
- Boost functionality
- NFT sale processing
- program-controlled liquidity

The intention is to move the important operating rules away from manually controlled project processes and into deterministic on-chain programs wherever practical.

In other words, fewer spreadsheets and fewer opportunities for somebody to press the wrong button.

4. Verifiable Random Rewards

The previous version of Eco Bitcoin modelled rewards as virtual mining output based upon miner speed and emissions.

The Solana version replaces this with weighted random reward draws.

Random selection requires a source of randomness that participants can independently trust.

Blockchain systems are deterministic by design, which is extremely useful until somebody asks one to generate an unpredictable number.

Using easily observable blockchain values such as timestamps or slots as pseudo-randomness can introduce opportunities for prediction or manipulation.

Eco Bitcoin therefore uses a Verifiable Random Function (VRF) / verifiable randomness process for reward selection.

Request → Randomness → Settlement

A participant submits an eligible NFT for a reward draw.

Randomness is generated and committed through the project's randomness provider.

Once the random value is available, the eBTC program deterministically calculates the result using the eligible NFT's draw weight and any applicable Boost.

This separates the request from the final random outcome and provides a result that can be independently verified rather than secretly selected by the project.

Which is important.

“Trust us, it was random” is not a particularly impressive cryptographic primitive.

5. POP NFT Tiers

The eBTC POP collection has a maximum supply of 4,269 NFTs.

Of these, 166 NFTs are reserved for migration of eligible legacy NFTs and 4,103 NFTs are available through the Solana public mint.

Legacy NFTs are migrated without requiring their holders to purchase the corresponding NFT again.

Tier	Price (SOL)	Approx. USD*	Max Supply	Legacy	Public Mint	Reward Multiplier	Draw Weight
FX100	0.100	$12.08	1,500	32	1,468	1.00×	10
FX250	0.250	$30.21	1,000	22	978	1.25×	25
GFX500	0.500	$60.42	900	22	878	1.50×	36
GFX750	0.750	$90.63	500	40	460	2.00×	60
GFX808	0.808	$97.64	369	50	319	3.00×	78
Total			**4,269**	**166**	**4,103**		

*USD values are illustrative values based upon the SOL price used when this table was prepared.

The collection therefore combines scarcity with progressively greater POP characteristics.

An FX100 provides the baseline participation level.

A GFX808 has a substantially higher weighting and reward multiplier but is also considerably scarcer, with only 369 capable of existing.

6. Draw Weight

Reward draws operate using weighted selection.

An NFT with a draw weight of 10 receives ten weighting units within the relevant draw.

An NFT with a draw weight of 78 receives seventy-eight.

$$\textbf{Probability Weight}_i = \textbf{Base Draw Weight}_i \times \textbf{Applicable Boost}_i$$

The probability of a particular eligible NFT being selected is therefore proportional to its effective weight compared with the combined effective weight of all NFTs participating in that draw.

$$P(i) = \text{EffectiveWeight}_i \,/\, \Sigma\ \text{EffectiveWeight}$$

This does not mean a GFX808 is guaranteed to win.

Randomness would be rather less interesting if the answer was known beforehand. It means the GFX808 has a greater probability of selection than NFTs with lower weighting.

The system therefore reproduces an idea familiar from mining: greater mining capability provides a greater statistical opportunity to receive a reward, without requiring anybody to actually mine anything.

7. Reward Multipliers

Draw Weight and Reward Multiplier perform different jobs.

Draw Weight affects the probability of winning.

Reward Multiplier affects the reward associated with a winning NFT.

The base multiplier is 1.00× for FX100.

- FX250 — 1.25×
- GFX500 — 1.50×
- GFX750 — 2.00×
- GFX808 — 3.00×

This gives the NFT tiers meaningful differences beyond simply increasing the odds of selection. It also means the most scarce POP NFTs represent both increased selection weight and increased potential reward.

Virtual mining without the noise.

8. Boost

Mining has always contained an element of competition. Eco Bitcoin's Boost feature adds a similar competitive element without requiring anybody to overclock anything.

A holder may activate a Boost for an eligible draw.

When successfully used, the Boost temporarily increases the NFT's effective draw weighting for that draw.

$$\textbf{Boosted Draw Weight} = \textbf{Base Draw Weight} \times 2$$

The Boost applies to the probability calculation for that draw only.

It does not permanently change the NFT's base tier.

It does not double the tier's reward multiplier.

And it does not guarantee a win.

For example, a GFX500 normally has a draw weight of 36. When Boosted for an eligible draw, 36 × 2 = 72. Its reward multiplier remains 1.50×.

The holder has therefore increased the NFT's chance of selection rather than simply purchasing a larger payout.

Boost introduces a gamification layer to POP and gives holders a mechanism through which participation can become more strategic than merely owning an NFT and periodically checking whether anything interesting happened.

The underlying VRF randomness remains unchanged.

A Boost improves the odds; it does not get to choose the number.

9. Legacy Migration

Eco Bitcoin existed on the Stellar network before its move to Solana.

Existing eligible participants should not therefore be expected to repurchase something merely because the project changed blockchain.

A total of 166 POP NFTs within the collection supply are allocated to verified legacy migration.

Tier	Legacy NFTs
FX100	32
FX250	22
GFX500	22
GFX750	40
GFX808	50
Total	**166**

Following verification through the migration process, the corresponding Solana NFT is issued to the legitimate holder. Migrated NFTs become normal members of the Solana POP collection and participate according to their respective tier characteristics.

There is no special “old NFT” penalty. Ageism is unnecessary, particularly for digital objects.

10. eBTC Supply and Emissions

The maximum supply of Eco Bitcoin remains 21,000,000 eBTC.

The Solana allocation of this supply is detailed in the Stellar to Solana Migration section, including the 16.8 million eBTC emission pool, 2.7 million eBTC liquidity allocation and 1.5 million eBTC migration reserve.

The number is not accidental and neither, admittedly, is much else about a project called Eco Bitcoin.

The intention remains to create a scarce digital asset whose distribution is progressively introduced through the project's participation and liquidity mechanisms rather than simply placing the entire supply into circulation at launch.

The original project used a scheduled virtual-mining emission model.

Under the Solana implementation, distribution is increasingly centred around POP reward draws and protocol liquidity.

Only eBTC entering circulation through the project's defined mechanisms contributes to liquid market supply.

The distinction between maximum supply and circulating supply is therefore important.

Having 21 million eBTC capable of existing does not mean 21 million eBTC are immediately available for sale.

Scarcity is considerably less convincing when everybody receives everything on day one.

11. Stellar to Solana Migration

Eco Bitcoin originally operated on the Stellar network. The move to Solana therefore required the existing eBTC supply, holder liabilities and liquidity to be accounted for rather than simply starting again with a fresh token allocation.

At the point of migration, the relevant Stellar eBTC position was:

Stellar eBTC allocation	eBTC
Not distributed	16,800,000
Holders migration liability	1,265,838
Liquidity pool	2,701,229

The Solana eBTC supply was then allocated as follows:

Solana eBTC allocation	eBTC
Distribution / emission pool	16,800,000
Liquidity pool	2,700,000
Migration reserve	1,500,000
Total	**21,000,000**

A migration reserve of 1,500,000 eBTC was created on Solana to cover eligible Stellar holder migration claims. The known Stellar holder migration liability at the time of this allocation was 1,265,838 eBTC.

Any eBTC remaining in the migration reserve after the migration process is complete will be added to protocol liquidity rather than retained by the team.

The Stellar liquidity position was also migrated economically rather than simply abandoned. Assets from the old Stellar liquidity pool were converted to USDC and subsequently into SOL. That SOL was then used to help seed the new Solana SOL/eBTC liquidity pool.

The migration therefore preserves the economic purpose of the original project while moving its operating infrastructure to Solana: the undistributed eBTC remains available for future emissions, existing holders retain a defined migration path, and legacy liquidity is carried forward into the new market structure.

Different chain. Same coins. Considerably more paperwork.

12. NFT Sales and Liquidity

The POP NFT sale mechanism is designed so that NFT issuance contributes directly to the eBTC economy and helps fund the infrastructure required to keep the system operating.

When a public POP NFT is sold, the SOL received is divided programmatically between the reward infrastructure and liquidity creation.

The current target allocation is:

- 5% — allocated to the keeper responsible for calling and settling reward draws;
- 40–45% — used to purchase eBTC from the market;
- the remaining 50–55% — retained as SOL for pairing with the purchased eBTC and adding liquidity.

POP NFT Sale

↓

SOL received by program

↓

5% allocated to Keeper / draw costs

↓

40–45% of sale proceeds used to purchase eBTC

↓

Purchased eBTC + remaining SOL

↓

Deposited into SOL/eBTC liquidity

The exact proportions used in an individual liquidity transaction may vary slightly because swaps, market prices and liquidity-pool ratios are not static.

The objective remains constant: NFT sales create eBTC demand, additional protocol liquidity and the funding required to keep the POP reward system operating.

The project team does not take income from Eco Bitcoin.

The intention is therefore for the protocol to be as self-funding as reasonably possible rather than relying upon the team to continually subsidise network fees and reward-draw infrastructure.

The remaining 95% of NFT sale proceeds are used by the liquidity mechanism rather than being retained as project revenue.

In other words, selling a POP NFT is intended to strengthen the system the NFT subsequently participates in.

13. Keeper and Reward Draw Funding

Eco Bitcoin reward draws require transactions to be submitted to the Solana network and randomness to be requested and settled.

Those operations have a real cost.

Current estimates place the annual cost of operating the reward-draw keeper at approximately 25–30 SOL per year.

The exact cost will vary according to network activity, draw frequency, transaction requirements and the cost of the underlying randomness service.

The project team receives no income from Eco Bitcoin and the system is intended to be self-funding.

For this reason, 5% of proceeds from public POP NFT sales are allocated to the keeper responsible for calling and completing reward draws.

The keeper allocation is intended to cover costs associated with:

- initiating reward draws
- requesting verifiable randomness
- submitting Solana transactions
- settling completed draws
- transaction fees
- keeper execution costs
- other directly related reward-system operating costs

The allocation is operational funding. It is not a team fee and it is not project income.

It exists because automated transactions still cost money, and asking the team to personally fund approximately 25–30 SOL of annual operating costs would make the system dependent upon continued manual subsidy.

That would rather defeat the point.

The objective is instead that activity within the POP ecosystem funds the infrastructure required to keep that ecosystem functioning.

The system should, as far as practical, pay for itself.

14. Program-Controlled Liquidity

Approximately 95% of public NFT sale proceeds are allocated to the eBTC liquidity mechanism.

A target 40–45% of the original NFT sale value is used to purchase eBTC from the market.

The acquired eBTC is then paired with the remaining SOL allocated to liquidity and deposited into the supported SOL/eBTC liquidity pool.

Because the market price of eBTC, swap execution and pool ratios can change between transactions, the final quantities deposited into the pool may differ slightly from the headline percentage allocation.

The important distinction is that these assets are not transferred to a conventional project-controlled treasury wallet.

The liquidity position is instead designed to be controlled by the program. There is therefore no ordinary private key held by the project that can simply withdraw the protocol-owned liquidity.

Swaps can continue through the decentralised exchange while liquidity remains subject to the program's rules.

This changes the trust model from:

"We promise not to remove the liquidity."

to something considerably more useful:

"The liquidity is controlled by the program rather than by our wallet."

The project team does not receive a percentage of NFT sales as income. The only portion removed from the liquidity mechanism is the 5% required to fund the keeper and reward infrastructure.

Crypto has historically provided enough examples demonstrating why distinguishing operational costs from project revenue is important.

15. A Self-Funding POP System

The Solana implementation is designed around a simple principle:

participation should help fund participation.

The project does not have a conventional revenue allocation for the team. The team purchased NFT's the same as everyone else, likewise with the eBTC it holds. The team's wallets are listed on the website for transparency.

Instead, public NFT sale proceeds are recycled into the mechanisms required for the project to continue operating.

POP NFT purchased

↓

5% funds Keeper / reward-draw operations

↓

40–45% of sale SOL buys eBTC

↓

Purchased eBTC + remaining liquidity SOL

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Added to SOL/eBTC liquidity

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POP NFT participates in VRF reward draws

↓

eBTC rewards distributed

↓

Participation continues

This creates three direct functions from a public POP NFT sale.

First, the NFT provides the holder with Proof of Participation.

Second, a small proportion of the sale contributes towards the approximately 25–30 SOL annual cost of operating the automated reward system.

Third, the overwhelming majority of the proceeds are recycled into eBTC purchases and protocol liquidity.

The intention is not to create a revenue stream for the project team.

It is to create a system capable of covering its own operating costs while directing the majority of new participant funds back into liquidity.

16. The eBTC Economic Loop

The Solana implementation creates a feedback mechanism between participation, token demand and liquidity.

The NFT sale therefore performs three functions simultaneously: it issues a new Proof of Participation asset, it contributes towards the ongoing cost of operating verifiable reward draws, and it creates market demand and liquidity for eBTC.

The intention is not to construct a perpetual-motion machine, those remain unavailable even on Solana.

The aim is simply to make the economics of NFT issuance contribute directly to the system the NFT participates in.

17. Scarcity

Eco Bitcoin contains several independent scarcity mechanisms.

eBTC maximum theoretical supply: 21,000,000 eBTC.

POP NFT maximum collection supply: 4,269 NFTs.

The higher POP tiers become progressively more limited:

- FX100 — 1,500
- FX250 — 1,000
- GFX500 — 900
- GFX750 — 500
- GFX808 — 369

NFTs cannot simply be issued indefinitely to increase reward participation.

Once the maximum collection supply has been reached, the POP supply has been reached.

This matters because draw weight derives part of its meaning from a finite participant set.

An NFT representing one of 369 possible GFX808 positions should remain one of 369 possible GFX808 positions.

Otherwise “maximum supply” becomes another crypto phrase meaning “maximum supply until we fancy changing it”.

18. Why POP?

Proof of Participation is intentionally simple.

Traditional mining asks: How much computational work have you contributed?

Proof of Stake broadly asks: How much economic stake have you committed?

Eco Bitcoin POP asks: Do you hold an eligible participation NFT, and what participation characteristics does it have?

The NFT becomes the portable cryptographic representation of participation.

It can be held.

It can be transferred.

It can be verified on-chain.

And because eligibility follows ownership, transferring the NFT also transfers its future POP utility to the new owner according to the rules of the program.

The NFT is therefore not intended to exist merely as a JPEG with aspirations.

It performs a function.

19. Verification Rather Than Trust

A central objective of the Solana redesign is to reduce reliance upon project-controlled processes.

Several important functions can be independently inspected or verified on-chain:

- NFT ownership
- NFT tier
- finite collection supply
- draw requests
- VRF/randomness fulfilment
- deterministic draw settlement
- Boost application
- eBTC transfers
- NFT sale transactions
- eBTC purchases
- liquidity deposits
- program-controlled liquidity authority

No blockchain system completely eliminates trust.

Smart contracts contain code, code is written by humans (and AI). The purpose is therefore not to claim that trust has magically vanished. It is to make the important rules observable, deterministic and increasingly difficult to change or bypass without detection.

20. Risks

Eco Bitcoin is experimental.

Participation involves the normal risks associated with cryptocurrency systems, including but not limited to:

- smart-contract/program vulnerabilities
- Solana network disruption
- decentralised exchange risk
- liquidity risk
- token price volatility
- SOL price volatility
- randomness provider dependencies
- NFT market volatility
- regulatory changes
- the possibility that nobody wants to buy eBTC

POP reward weighting represents a probability.

It does not represent guaranteed income.

NFT ownership does not guarantee that the holder will receive rewards of any particular value or frequency.

A higher draw weight improves relative probability.

It does not negotiate with mathematics.

21. Future Work

Eco Bitcoin remains an evolving experiment.

Future development may include additional decentralisation of program authorities, expanded on-chain governance, further Boost mechanics, additional participation features and refinement of the reward system as real-world POP data becomes available.

The guiding principle remains that new functionality should strengthen participation or decentralisation rather than add complexity solely because blockchains allow remarkably complicated things to be built.

Sometimes not building something is also innovation.

22. Conclusion

Bitcoin demonstrated that digital scarcity can have value and that economic incentives can coordinate participants across a decentralised network.

It also demonstrated that humans, when offered a financial incentive to perform calculations, are capable of constructing industrial facilities containing extraordinary numbers of computers purely to guess numbers very quickly.

Eco Bitcoin explores a different question.

Can some of the economic ideas associated with mining — scarcity, probability, participation, competition and reward — exist without the physical mining process?

On Solana, Eco Bitcoin represents participation through a limited collection of functional POP NFTs.

Reward selection uses verifiable randomness.

NFT tiers establish different probabilities and reward characteristics.

Boost adds strategy and gamification.

NFT sales create eBTC demand and SOL/eBTC liquidity.

A 5% keeper allocation helps fund the approximately 25–30 SOL annual operating cost of calling and settling draws, allowing the project to remain self-funding without creating team income.

And that liquidity is controlled by the program rather than somebody's hardware wallet sitting in a desk drawer.

It is not Bitcoin.

It does not pretend to be Bitcoin.

It is an experiment inspired by Bitcoin.

A virtually mined, Proof of Participation interpretation of an idea that began with one rather influential whitepaper.

Only this time, hopefully, with considerably lower electricity bills.